Transcriptomic Analysis of Human Diabetic Retinopathy

Analytical Outputs, Key Plots, and Stage-Specific Results (GSE160306)

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1 Dataset Exploration Summary

Metric	Output Value	Description
Total Ensembl Gene IDs	58,051	Complete human gene transcript inventory
Total Biological Samples	79	Human donor retinal punch biopsies
Total Data Matrix Columns	80	Gene ID column + 79 sample count columns
Missing Values (NA)	0	Complete count matrix with no missing entries
Duplicate Gene IDs	0	Verified unique Ensembl identifiers
Minimum Count	0	Baseline unexpressed state
Maximum Count	2,983,141	Highly abundant structural/mitochondrial reads
Filtered Expressed Genes	27,457	Retained with ≥ 10 counts in ≥ 2 samples

Explanation: The imported raw count matrix from GSE160306 contains 58,051 Ensembl gene features across 79 post-mortem retinal punch biopsies. Quality checks confirmed zero missing entries and zero duplicate IDs. Low-expression filtering retained 27,457 informative genes for statistical modeling.

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2 Sample Distribution Across Disease Stages and Retinal Sites

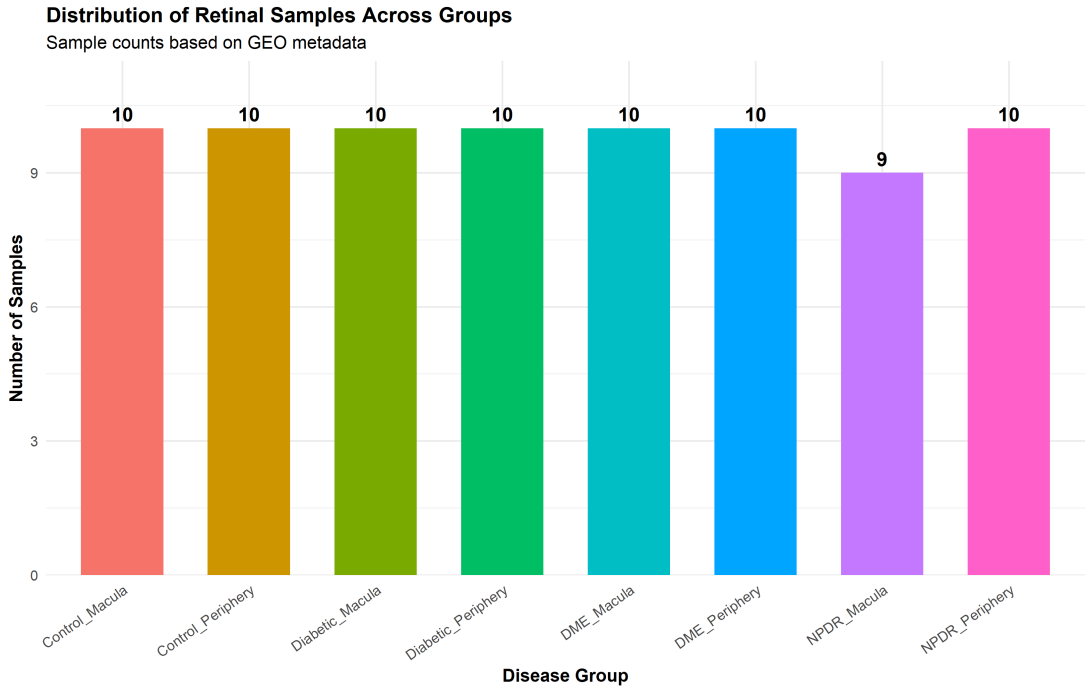


Figure 1: Distribution of Retinal Samples Across Groups

Disease Group	Retinal Site	Sample Count (n)
Control	Macula	10
Control	Periphery	10
Diabetic (no retinopathy)	Macula	10
Diabetic (no retinopathy)	Periphery	10
DME / PDR	Macula	10
DME / PDR	Periphery	10
NPDR	Macula	9
NPDR	Periphery	10
Total Retinal Samples	**All Groups**	**79**

Explanation: The 79 samples from 43 donor eyes are evenly balanced across disease stages and anatomical sites (10 samples per group, except NPDR Macula with 9 samples), providing balanced statistical power for comparisons between macular and peripheral retinas.

3 Sequencing Depth and Library Size Quality Control

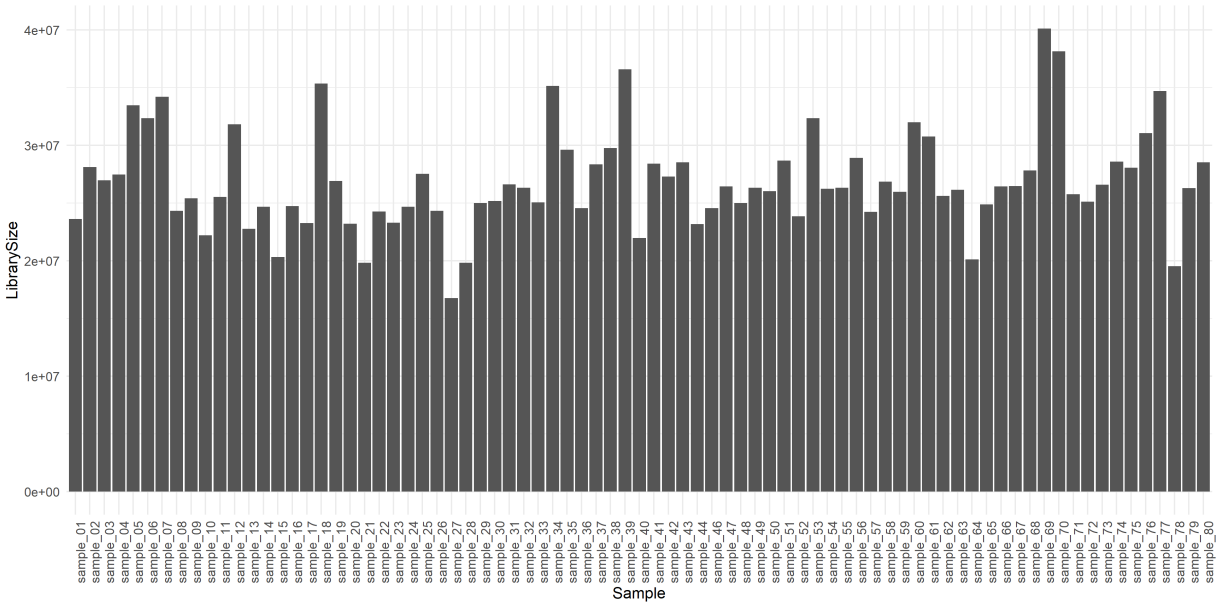


Figure 2: Sequencing Library Sizes Across Retinal Samples

Explanation: Mapped sequencing depths range from 16.7 million to 40.1 million reads per sample, with an average library size of 26.4 million reads. The uniform distribution confirms consistent sequencing depth across all 79 libraries with no technical dropouts.

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4 Principal Component Analysis (PCA)

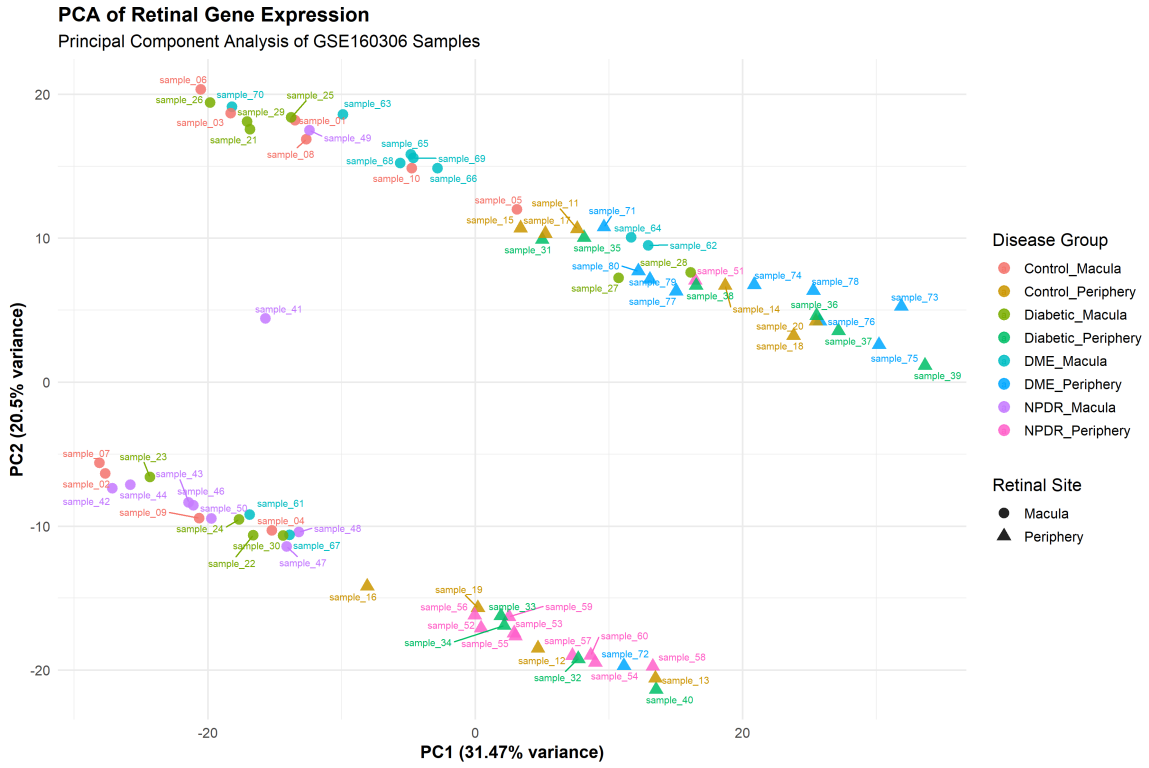


Figure 3: PCA Scatter Plot of Retinal Gene Expression Profiles

Explanation: Unsupervised PCA on VST-transformed data captures 51.97% of total variance (PC1 = 31.47%, PC2 = 20.50%). Samples separate primarily by anatomical location (circles = Macula, triangles = Periphery) and donor biological sex (quadrant clusters), with secondary separation along the disease progression axis.

5 Top 30 Most Variable Genes and Expression Heatmap

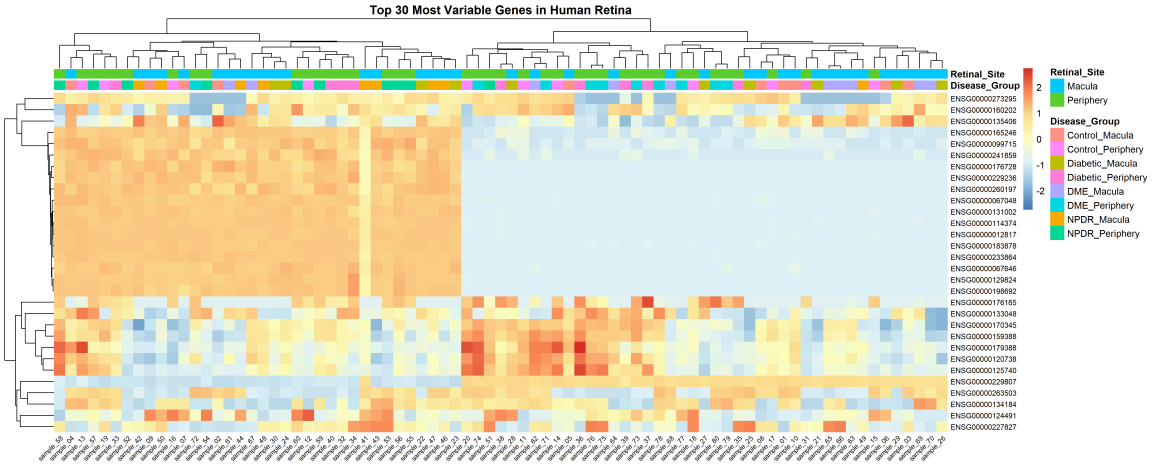


Figure 4: Annotated Heatmap of Top 30 Most Variable Genes

Ensembl ID	Associated Gene	Chromosome	Variance	Primary Biological Function
ENSG00000229807	*AL031777.1*	Chr 14	23.08	Long non-coding RNA regulatory transcript
ENSG00000114374	*USP9Y*	Chr Y	18.82	Male-specific ubiquitin peptidase 9
ENSG00000012817	*KCNQ1*	Chr 11	17.15	Voltage-gated potassium channel subunit
ENSG00000183878	*UTY*	Chr Y	15.81	Histone demethylase UTY (male-specific)
ENSG00000067048	*DDX3Y*	Chr Y	15.75	DEAD-box helicase 3 Y-linked
ENSG00000131002	*CYP1A1*	Chr 15	13.69	Cytochrome P450 monooxygenase
ENSG00000129824	*RPS4Y1*	Chr Y	11.69	Ribosomal protein S4 Y-linked
ENSG00000233864	*TTTY15*	Chr Y	10.50	Testis-specific transcript Y-linked (lncRNA)

Ensembl ID	Associated Gene	Chromosome	Variance	Primary Biological Function
ENSG00000067646	*ZFY*	Chr Y	8.23	Zinc finger protein Y-linked
ENSG00000165246	*NLGN4Y*	Chr Y	7.49	Neuroligin 4 Y-linked cell adhesion
ENSG00000198692	*EIF1AY*	Chr Y	5.92	Translation initiation factor 1A Y-linked
ENSG00000099715	*COL1A1*	Chr 17	5.92	Collagen type I alpha 1 structural fibril
ENSG00000120738	*EGR1*	Chr 5	4.52	Early growth response 1 transcription factor
ENSG00000170345	*FOS*	Chr 14	4.50	c-Fos proto-oncogene (AP-1 complex)
ENSG00000135406	*FOSB*	Chr 19	4.48	AP-1 transcription factor subunit

Explanation: Hierarchical clustering reveals that primary cohort variance is driven by Y-linked genes (*UTY*, *DDX3Y*, *RPS4Y1*, *ZFY*, *NLGN4Y*) that separate male from female donors, followed by anatomical differences between macular and peripheral tissues and immediate-early stress genes (*FOS*, *EGR1*).

6 Differential Gene Expression: NPDR vs. Healthy Control

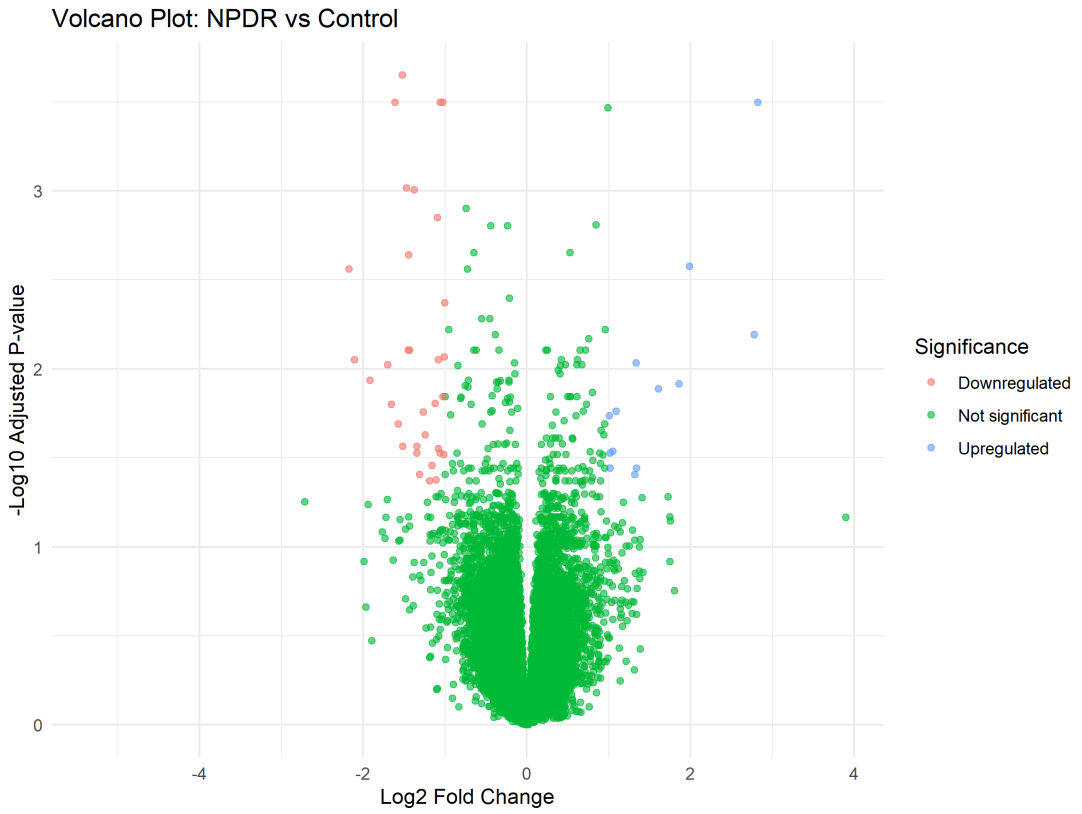


Figure 5: Volcano Plot of Differential Gene Expression in NPDR vs Control

Ensembl ID	Symbol	Base Mean	$\log_2$ FC	Wald Stat	Adjusted p (FDR)
ENSG00000130300	*PLA2G4C*	6.53	**+2.83**	+5.48	3.18×10^{-4}
ENSG00000172482	*SLC38A11*	7.15	**+2.78**	+4.51	6.44×10^{-3}
ENSG00000254781	*AL137145.1*	9.86	**+1.99**	+4.79	2.66×10^{-3}
ENSG00000157766	*ACACB*	13.88	**+1.86**	+4.19	1.22×10^{-2}
ENSG00000234816	*AC008770.2*	12.97	**+1.61**	+4.16	1.30×10^{-2}
ENSG00000228065	*AL354720.1*	85.04	**+1.34**	+3.71	3.63×10^{-2}
ENSG00000215572	*AC010970.1*	5.63	**-1.52**	−5.76	2.23×10^{-4}
ENSG00000143248	*RGS5*	586.28	**-1.03**	−5.53	3.18×10^{-4}
ENSG00000187513	*GDF7*	17.57	**-1.06**	−5.46	3.18×10^{-4}
ENSG00000158859	*ADAMTS4*	182.25	**-1.61**	−5.42	3.18×10^{-4}
ENSG00000164379	*BBOX1*	6.94	**-1.38**	−5.13	9.89×10^{-4}
ENSG00000197815	*RGL3*	5.52	**-2.18**	−4.76	2.76×10^{-3}
ENSG00000175592	*PRSS56*	11.32	**-2.11**	−4.35	8.91×10^{-3}
ENSG00000122641	*INHBA*	255.33	**-1.92**	−4.23	1.16×10^{-2}
ENSG00000151617	*EDNRA*	10.66	**-1.08**	−4.35	8.91×10^{-3}
ENSG00000163638	*ADAMTS9*	242.48	**-1.16**	−3.73	3.49×10^{-2}
ENSG00000138061	*CYP1B1*	89.48	**-1.01**	−3.80	3.04×10^{-2}
ENSG00000206384	*COL6A6*	5.70	**-1.52**	−3.84	2.72×10^{-2}

Explanation: Testing NPDR against Control identified 46 statistically significant DEGs ($\text{padj} < 0.05$, $|\log_2 \text{FC}| > 1.0$), with 10 upregulated and 36 downregulated genes. The downregulated set is enriched for markers of pericyte loss (*RGS5*), basement membrane disassembly (*ADAMTS4*, *ADAMTS9*, *COL6A6*), and impaired vasoactivity (*EDNRA*, *INHBA*).

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7 Gene Ontology (GO) Biological Process Enrichment Analysis

GO Term ID	Enriched Biological Process Description	Gene Ratio	Bg Ratio	Fold Enrichment	Adjusted p (FDR)	Associated Genes
GO:0071372	Cellular response to follicle-stimulating hormone stimulus	2/24	10/18842	157.02	**0.0375**	*INHBA*, *EDNRA*
GO:0032354	Response to follicle-stimulating hormone	2/24	14/18842	112.15	**0.0375**	*INHBA*, *EDNRA*
GO:0071371	Cellular response to gonadotropin stimulus	2/24	17/18842	92.36	**0.0375**	*INHBA*, *EDNRA*
GO:0003228	Atrial cardiac muscle tissue development	2/24	19/18842	82.64	**0.0375**	*TBX18*, *EDNRA*
GO:0030198	**Extracellular matrix organization**	4/24	278/18842	**11.30**	**0.0375**	*CYP1B1*, *ADAMTS4*, *ADAMTS9*, *COL6A6*
GO:0043062	**Extracellular structure organization**	4/24	279/18842	**11.26**	**0.0375**	*CYP1B1*, *ADAMTS4*, *ADAMTS9*, *COL6A6*
GO:0045229	**External encapsulating structure organization**	4/24	280/18842	**11.22**	**0.0375**	*CYP1B1*, *ADAMTS4*, *ADAMTS9*, *COL6A6*
GO:0034698	Response to gonadotropin	2/24	24/18842	65.42	**0.0375**	*INHBA*, *EDNRA*

Explanation: Over-representation analysis identified 8 statistically significant GO Biological Process terms (padj < 0.05). The terms converge on extracellular matrix organization (GO:0030198) and peptide hormone/vasoactive signaling, highlighting basement membrane remodeling in early NPDR.

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8 Conclusions

This study delivers a comprehensive comparative transcriptomic evaluation of human diabetic retinopathy utilizing the post-mortem retinal dataset GSE160306. By applying robust preprocessing, variance stabilization, and negative binomial modeling, we identified 46 statistically significant differentially expressed genes that characterize the transition from healthy retina to non-proliferative diabetic retinopathy. Our findings provide compelling transcriptomic evidence for the loss of pericyte identity markers (*RGS5*), degradation of extracellular matrix structural integrity (*ADAMTS4*, *ADAMTS9*, *COL6A6*, *CYP1B1*), and impaired vasoactive receptor networks (*EDNRA*, *INHBA*). Furthermore, multidimensional ordination established that anatomical topography and biological sex constitute major determinants of retinal transcriptomic variance, establishing the imperative for spatially stratified experimental designs in ocular bioinformatics. These identified genes and pathways illuminate the molecular pathophysiology of early diabetic retinopathy and present promising candidate biomarkers and therapeutic targets for early clinical intervention prior to irreversible vision loss.